

Lux: Technical Architecture Specification

Visual Output Surface for AI Agents

May 2026 — v0.18.0

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1 Overview

Lux is a visual output surface for Claude Code. It renders element trees pushed by AI agents via MCP tools into a native ImGui window over Unix socket IPC. The system has four projection surfaces from a single codebase: Python library, CLI, MCP server, and Claude Code plugin.

The display server owns the render loop. Agents are clients. All agent–display communication is message-passing over a framed JSON protocol on a Unix domain socket. The display server is single-threaded: no threads, no asyncio, no locks. It polls all I/O non-blocking each frame via `select()`.

1.1 Design Principles

- P1. Composable element tree.** Every element kind is small, nestable, and data-driven. If an agent can describe it as JSON, Lux renders it.
- P2. Protocol is the API surface.** The wire protocol is the only interface between agents and the display. No shared memory, no function calls, no Python imports required.
- P3. Single-threaded simplicity.** The display server uses a polling event loop with zero threads. All state is owned by the main thread. No synchronization primitives.
- P4. Incremental by default.** Scenes can be patched by element ID without replacing the entire tree. Table filters run at frame rate with zero MCP round-trips.
- P5. Multi-scene coexistence.** Each `show()` call opens a new tab. Scenes persist until explicitly dismissed or cleared. Same `scene_id` replaces content in-place.

2 System Architecture

2.1 Component Map

Module	File	Responsibility
Protocol	<code>src/punt_lux/protocol.py</code>	Message types, element dataclasses, wire format (framing, serialization, deserialization)
Display	<code>src/punt_lux/display.py</code>	ImGui render loop, socket IPC, scene state, widget state, element renderers, event flush
Client	<code>src/punt_lux/client.py</code>	LuxClient context manager; connects, sends, receives, buffers pending messages
Display Client	<code>src/punt_lux/display_client.py</code>	High-level client with typed methods for show, update, introspect, screenshot
MCP Tools	<code>src/punt_lux/tools.py</code>	FastMCP tools (24 tools: show, update, clear, ping, recv, introspection, configuration); auto-reconnect
Hub	<code>src/punt_lux/hub.py</code>	WebSocket session hub; multiplexes MCP sessions, routes scene updates to display
Show	<code>src/punt_lux/show.py</code>	CLI subcommands for pre-built scenes (beads board)
Apps/Beads	<code>src/punt_lux/apps/beads.py</code>	Beads issue browser; loads <code>.beads/issues.jsonl</code> , renders filterable table
Paths	<code>src/punt_lux/paths.py</code>	Socket discovery, PID lifecycle, auto-spawn
Runtime	<code>src/punt_lux/runtime.py</code>	CodeExecutor ; compiles and runs render functions per-frame
Config	<code>src/punt_lux/config.py</code>	<code>.lux/config.md</code> YAML frontmatter; display mode toggle
Hooks	<code>src/punt_lux/hooks.py</code>	Claude Code lifecycle hook handlers
Service	<code>src/punt_lux/service.py</code>	Daemon lifecycle management (start, stop, status)
Remote	<code>src/punt_lux/remote.py</code>	mcp-proxy configuration and WebSocket transport setup
CLI	<code>src/punt_lux/_main_.py</code>	Typer CLI: display, serve, enable, disable, status, doctor, install, uninstall, hub-install, hub-uninstall, hub-status, ensure-hub, setup-proxy, show, ping, version

2.2 Data Flow

Three primary data paths connect agents to the display:

2.2.1 Write Path (Agent → Display)

1. Agent calls MCP tool (e.g., `show()`).
2. MCP server creates **LuxClient**, calls `client.show(scene_id, elements)`.
3. Client serializes **SceneMessage** as JSON, frames with 4-byte big-endian length prefix, sends over Unix socket.
4. Display receives frame in `_poll_clients()`, deserializes, stores in `_scenes` dict.
5. Display renders scene each frame via `_render_scene()`.
6. Display sends **AckMessage** back to client.

2.2.2 Event Path (User → Agent)

1. User interacts with widget (click, slider drag, text input).
2. Display generates `InteractionMessage` with `element_id`, `action`, `value`, and optional `scene_id`.
3. Message buffered in `_event_queue`.
4. Each frame, `_flush_events()` broadcasts all queued messages to all connected clients, then clears the queue.
5. Agent calls `recv()` to consume events.

2.2.3 Update Path (Incremental Patching)

1. Agent calls `update(scene_id, patches)`.
2. Display receives `UpdateMessage`, calls `_apply_update()`.
3. `_find_element()` locates target by ID via depth-first search through the element tree.
4. Patch applied: `set` updates fields in-place, `remove` deletes element and cleans up widget state.
5. Routes to correct scene by `scene_id` (not limited to active tab).

3 Wire Protocol

3.1 Framing

Every message is a length-prefixed JSON frame:

```
+-----+-----+
| 4 bytes (uint32) | N bytes (UTF-8 JSON) |
| big-endian | payload |
+-----+-----+
```

Maximum payload size: 16 MiB (2^{24} bytes). The `FrameReader` accumulates partial reads into a byte buffer and yields complete frames. Oversize frames trigger client disconnection.

3.2 Message Types

3.2.1 Client → Display

Type	Purpose
<code>SceneMessage</code>	Replace or add scene. Fields: <code>id</code> , <code>elements[]</code> , <code>title</code> , <code>layout</code> .
<code>UpdateMessage</code>	Incremental patches. Fields: <code>scene_id</code> , <code>patches[]</code> (each: <code>id</code> , <code>set</code> , <code>remove</code>).
<code>ClearMessage</code>	Remove all scenes and reset state.
<code>PingMessage</code>	Heartbeat. Field: <code>ts</code> .
<code>MenuMessage</code>	Set agent-provided menu entries.
<code>ThemeMessage</code>	Apply theme by name.
<code>ConnectMessage</code>	Client identity handshake. Field: <code>name</code> .
<code>RegisterMenuMessage</code>	Register per-client menu items. Field: <code>items[]</code> .
<code>IntrospectRequest</code>	Request scene element tree. Field: <code>scene_id</code> .
<code>ListScenesRequest</code>	Request list of all active scenes.
<code>ScreenshotRequest</code>	Request display screenshot.
<code>QueryRequest</code>	Generic introspection query. Field: <code>query_type</code> .

SceneMessage frame fields (since v0.9.0): `SceneMessage` includes optional fields `frame_id`, `frame_title`, `frame_size` (initial `[width, height]`), `frame_flags` (dict of ImGui window flags such as `no_resize`, `no_collapse`, `no_title_bar`, `no_background`, `no_scrollbar`, `auto_resize`), and `frame_layout` ("tab" or "stack").

3.2.2 Display → Client

Type	Purpose
ReadyMessage	Handshake on connect; protocol version, capabilities.
AckMessage	Acknowledges scene or update; <code>scene_id</code> , <code>ts</code> , optional <code>error</code> .
InteractionMessage	User interaction; <code>element_id</code> , <code>action</code> , <code>value</code> , <code>ts</code> , optional <code>scene_id</code> .
PongMessage	Ping response; round-trip latency probe.
IntrospectResponse	Scene element tree response.
ListScenesResponse	List of active scene summaries.
ScreenshotResponse	Base64-encoded display screenshot.
QueryResponse	Generic introspection response.

Unknown message types are deserialized as `UnknownMessage` for forward compatibility rather than disconnecting the client.

3.3 Element Kinds

24 element kinds organized by category:

Category	Element	Notes
Text	<code>text</code>	Content + style (heading, caption, success, error, code); optional <code>color</code> hex string
	<code>markdown</code>	CommonMark rendering via <code>imgui_md</code>
Media	<code>image</code>	File path or base64 data
Interactive	<code>button</code>	Label, action, disabled flag; <code>arrow</code> (directional) and <code>small</code> variants
	<code>slider</code>	Min/max/value, integer mode
	<code>checkbox</code>	Boolean toggle
	<code>combo</code>	Dropdown selection
	<code>input_text</code>	Single-line text input
	<code>radio</code>	Radio button group
	<code>input_number</code>	Numeric input with step buttons, min/-max clamping, integer mode
Data	<code>color_picker</code>	RGB/RGBA hex picker; <code>alpha</code> (RGBA) and <code>picker</code> (full widget) modes
	<code>table</code>	Columns, rows, built-in filters, detail panel, pagination
	<code>plot</code>	ImPlot: line, scatter, bar series
	<code>progress</code>	Fraction bar with label
Lists	<code>spinner</code>	Loading indicator
	<code>selectable</code>	Click-to-select item
Drawing	<code>tree</code>	Recursive node hierarchy; <code>flat</code> flag for non-indented expansion
	<code>draw</code>	Canvas with draw commands
Layout	<code>modal</code>	Modal popup dialog; container with children, emits <code>"closed"</code> event
	<code>group</code>	Row, column, or paged layout (<code>pages</code> + <code>page_source</code>)
	<code>tab_bar</code>	Tabbed container
	<code>collapsing_header</code>	Expandable section

Category	Element	Notes
	window	Movable, resizable floating panel
	separator	Visual divider

4 Display Server Internals

4.1 Frame Cycle

The display server runs an ImGui render loop via `immapp.run()`. Each frame executes four phases in sequence:

1. **Accept connections.** `select()` on server socket with timeout 0. Accept if readable. Send `ReadyMessage` to new client.
2. **Poll clients.** `select()` on all client sockets with timeout 0. Read available data into per-client `FrameReader`. Dispatch complete messages to `_handle_message()`. Remove errored sockets.
3. **Render scene.** Enable docking via `dock_space_over_viewport`. Render idle flame background. Check for World menu background click. Render World panel if open. Render all workspace frames (see §5). Render dock bar for minimized frames. Legacy fallback: if no frames, render unframed scenes as tabs.
4. **Flush events.** Broadcast all buffered `InteractionMessages` to all connected clients, then clear the queue.

All four phases are non-blocking. The frame completes in microseconds when idle. ImGui handles frame pacing via GLFW vsync.

4.2 Multi-Scene State

Field	Type and Purpose
<code>_scenes</code>	<code>dict[str, SceneMessage]</code> — ordered by insertion; keyed by <code>scene_id</code> .
<code>_scene_order</code>	<code>list[str]</code> — explicit tab ordering.
<code>_active_tab</code>	<code>str None</code> — currently selected tab.
<code>_scene.widget_state</code>	<code>dict[str, WidgetState]</code> — per-scene slider/checkbox/combo/text state.
<code>_event_queue</code>	<code>list[InteractionMessage]</code> — global, shared across all scenes.
<code>_dirty_windows</code>	<code>set[str]</code> — window element IDs needing initial position set.
Frame state (since v0.9.0)	
<code>_frames</code>	<code>dict[str, _Frame]</code> — all workspace frames, keyed by <code>frame_id</code> .
<code>_focus_frame_id</code>	<code>str None</code> — frame to auto-focus on next render.
<code>_scene.to_frame</code>	<code>dict[str, str]</code> — scene → frame lookup.
<code>_scene.to_owner</code>	<code>dict[str, int]</code> — scene → client fd ownership.
Client identity (since v0.9.0)	
<code>_client_names</code>	<code>dict[int, str]</code> — client fd → display name.
<code>_menu_registrations</code>	<code>dict[int, list]</code> — per-client menu items.
<code>_menu_owners</code>	<code>dict[str, int]</code> — menu item ID → owner fd.

4.2.1 Scene Lifecycle

New scene (`scene_id` not in `_scenes`)

Appended to `_scenes` and `_scene_order`. Fresh `WidgetState` allocated. Becomes the active tab. Window elements marked dirty. Existing events preserved.

Replace-in-place (same `scene_id`)

Content replaced in `_scenes`. Widget state and render function state cleared for that scene. Events for elements that existed in the old scene but not the new scene are drained (using full subtree ID collection via `_collect_ids()`). Events from other scenes are untouched.

Dismiss (tab close button)

Scene removed from all state dicts. Window element IDs removed from `_dirty_windows`. Active tab moves to the adjacent neighbor (next tab, or previous if rightmost was dismissed).

Clear (ClearMessage or menu)

All scenes, widget state, events, and dirty windows cleared. Fresh instances allocated (not `.clear()` on aliased objects).

4.2.2 Widget State Swap Pattern

The 20+ element renderers all read from `self._widget_state`. Rather than threading a `scene_id` parameter through every method, the display swaps the per-scene state dict into this instance variable before rendering each tab:

```
def _render_scene_tab(self, scene_id: str) -> None:
    self._widget_state = self._scene_widget_state[scene_id]
    scene = self._scenes[scene_id]
    for elem in scene.elements:
        self._render_element(elem)
```

This is the same pattern ImGui itself uses: “current context.” Zero changes required in any renderer method.

4.3 Table Filtering

Tables support built-in filtering at frame rate with zero MCP round-trips. The display server renders filter controls (search input, combo dropdowns) above the table and applies filters in the render loop:

1. Filter state stored in `WidgetState` with prefixed keys (e.g., `_tbl_search_{eid}`, `_tbl_filter_{col}_{eid}`).
2. `_apply_table_filters()` renders controls, snapshots state, detects changes.
3. `_filter_indexed_rows()` applies AND logic across all active filters (case-insensitive substring for search, exact match for combos).
4. Pagination resets to page 0 when filters change.
5. Detail panel renders a two-column metadata grid plus long-form body text for the selected row.

5 Frame Architecture

Since v0.9.0, scenes can target named *frames*—persistent ImGui windows that outlive client connections. Frames are the primary unit of workspace organization.

5.1 Frame Dataclass

Field	Type and Purpose
<code>frame_id</code>	<code>str</code> — Unique identifier.
<code>title</code>	<code>str</code> — Window title bar text.
<code>owner_fds</code>	<code>set[int]</code> — File descriptors of contributing clients.
<code>scenes</code>	<code>dict[str, SceneMessage]</code> — All scenes in this frame, keyed by <code>scene_id</code> .
<code>scene_order</code>	<code>list[str]</code> — Tab ordering.
<code>active_tab</code>	<code>str</code> <code>None</code> — Currently visible scene.
<code>minimized</code>	<code>bool</code> — Docked to bottom bar.
<code>cascade_index</code>	<code>int</code> — Stagger offset for initial positioning.
<code>initial_size</code>	<code>tuple[int,int]</code> <code>None</code> — Width/height hint (first use only).
<code>flags</code>	<code>dict[str,bool]</code> <code>None</code> — ImGui window flags (<code>no_resize</code> , <code>no_collapse</code> , <code>no_title_bar</code> , <code>no_background</code> , <code>no_scrollbar</code> , <code>auto_resize</code>).
<code>layout</code>	<code>"tab"</code> <code>"stack"</code> — Multi-scene composition mode.

5.2 Frame Lifecycle

Creation

A frame is created on the first `SceneMessage` with a `frame_id` not already in `_frames`. The client's fd is added to `owner_fds`.

Shared ownership

Multiple clients can contribute scenes to the same frame. Each client that sends a scene targeting the frame is added to `owner_fds`.

Orphan model

When a client disconnects, its scenes persist with an `_ORPHAN_FD` sentinel in place of the real fd. The frame remains visible and interactive. A new client connecting with the same `frame_id` adopts the orphaned scenes.

Dismissal

The close button removes the frame and all its scenes from state. Widget state for all scenes in the frame is cleaned up.

Focus management

New frames are positioned using cascade staggering (`cascade_index` increments per frame). Initial size is applied on first render only. `_focus_frame_id` triggers `SetNextWindowFocus` on the next render cycle.

5.3 World Menu

Right-clicking the docking background opens the World menu panel, providing frame management operations: list all frames, restore minimized frames, close all frames, and access display settings.

5.4 Dock Bar

Minimized frames appear as clickable buttons in a bottom dock bar. Clicking a dock bar button restores the frame to its previous position and size.

6 IPC and Process Management

6.1 Socket Discovery

Socket path resolution, in priority order:

1. `$LUX_SOCKET` environment variable.

2. `$XDG_RUNTIME_DIR/lux/display.sock`.
3. `/tmp/lux-$USER/display.sock` (fallback).

The fallback path is chosen to stay within macOS’s ~104-character `AF_UNIX` path limit.

6.2 PID Lifecycle

- PID file written at `{socket_path}.pid` on server startup.
- Clients verify liveness via `os.kill(pid, 0)` before connecting.
- Stale socket cleanup: `cleanup_stale_socket()` removes socket and PID file if the process is dead.

6.3 Auto-Spawn

`ensure_display()` spawns the display server if not running:

1. Checks socket existence and PID liveness.
2. If not running, spawns


```
python -m punt_lux display --socket {path}
with start_new_session=True
```

 (detached from caller’s process group).
3. Polls socket existence + PID liveness every 100 ms until ready or 5 s timeout.
4. Display logs to `{socket_path}.log`.

6.4 MCP Server Reconnect

The MCP server wraps every tool call in `_with_reconnect()`: if an `OSError` (broken pipe) occurs, the stale client is closed, a new connection is established, and the call is retried exactly once. No retry loop.

7 Performance

7.1 Frame Budget

Parameter	Value	Notes
Idle FPS	30	<code>fps_idling.fps_idle = 30.0</code>
Active FPS	60+	GLFW vsync, GPU-bound
Socket poll timeout	0	Non-blocking <code>select()</code>
Max message size	16 MiB	Hard limit in <code>FrameReader</code>
Connection timeout	5.0 s	<code>ensure_display()</code> and handshake
Texture loading	Lazy	First render triggers PIL + OpenGL upload

7.2 Critical Path Analysis

The frame-critical path is:

1. `select()` on N client sockets — $O(N)$, $N < 64$ in practice.
2. `FrameReader.feed()` — memcpy into buffer, yield frames. $O(B)$ where B is bytes received.
3. `_render_scene()` — ImGui draw list calls. $O(E)$ where E is total element count across visible scene.
4. `_flush_events()` — send K events to N clients. $O(K \cdot N)$.

Table filtering is $O(R \cdot C)$ where R is row count and C is column count. This runs every frame but is fast for tables under ~10,000 rows. No benchmarks exist for larger tables; pagination (configurable page size) bounds visible row rendering.

7.3 Idle Screen

The idle screen renders ~60 draw calls per frame: 48 radial lines, 3 glow circles, 3 flame shapes (each ~7 bezier operations). This is negligible for a GPU-rendered immediate-mode UI at 30 FPS idle.

7.4 Memory

- **Scene storage.** Scenes are Python dataclass trees held in memory. No size limit beyond the 16 MiB message cap. Each scene is a separate dict entry; dismissed scenes are garbage-collected.
- **Texture cache.** OpenGL texture IDs held until `_on_exit()`. No eviction policy. Image-heavy scenes accumulate GPU memory.
- **Widget state.** Per-scene `WidgetState` is a flat dict. Cleared on scene replacement. No growth concern.
- **Event queue.** Flushed every frame. Bounded by interaction rate (<100 events/frame in practice).

7.5 Known Limitations

1. **No ImGui ID scoping.** Widget IDs are derived from `element_id` alone, not scoped by scene. ImGui internal state (focus, open/closed) can bleed between tabs with colliding IDs.
2. **No texture eviction.** Long-running sessions with many images accumulate GPU memory.
3. **Filter is per-frame.** Table filtering re-executes every frame even when nothing changed. Not a problem at current scale but would benefit from dirty-flag optimization for 10,000+ row tables.

8 Security

8.1 Threat Model

The primary trust boundary is the Unix socket. The display server accepts connections from any process that can reach the socket file. In the default configuration (`/tmp/lux-$USER/`), the socket is protected by filesystem permissions (user-only directory).

Threat	Description	Mitigation
Socket hijacking	Another process connects to the socket and injects scenes or reads events.	Directory permissions (0700) on <code>/tmp/lux-\$USER/</code> .
Malformed messages	Client sends invalid JSON, unknown message type, or oversize payload.	Unknown types deserialized as <code>UnknownMessage</code> ; oversize payloads rejected by <code>FrameReader</code> .
Render function	Agent-submitted Python code executes arbitrary operations.	Consent dialog with AST warnings. User must explicitly click Allow.
Resource exhaustion	Scene with millions of elements or very large images.	16 MiB message cap. No element count limit (ImGui handles gracefully).

8.2 Render Function Security

Render functions are the only element kind that executes arbitrary code. The security model is **consent-based, not sandbox-based**:

1. **AST scanning:** a best-effort visitor flags suspicious imports (`os`, `subprocess`, `socket`, `ctypes`, `urllib`, etc.) and dangerous builtins (`eval`, `exec`, `open`, `__import__`). This is a *UX signal*, not a security boundary.

2. **Consent dialog:** an ImGui modal displays the full source code with line numbers and any AST warnings. The user must click “Allow” to proceed. Dismissing the modal (Esc) is treated as “Deny.” Consent is per-element: changing the source code resets consent.
3. **Execution:** `CodeExecutor` compiles the source and calls `render(ctx)` each frame. The `RenderContext` provides a persistent state dict, delta time, frame counter, canvas dimensions, and a `send()` call-back for emitting events. Execution is **not sandboxed**: the code runs in the display server’s process with full access to the Python runtime. Runtime errors are caught and displayed, not propagated.

Implication: render functions should be treated as user-approved code, equivalent to running a script the user downloaded. The consent dialog provides informed consent, not isolation.

8.3 Protocol Robustness

- **Malformed JSON:** client disconnected, not crashed. The display logs the error and removes the client from `_clients`.
- **Unknown message type:** deserialized as `UnknownMessage` for forward compatibility.
- **Oversize frame:** `FrameReader` checks `bufSize ≤ maxBufSize` and disconnects on violation.
- **Partial reads:** `FrameReader` accumulates bytes across frames; no assumption of complete messages per `recv()`.
- **Identity fields protected:** `_apply_update()` silently drops patches that attempt to change `id` or `kind` on an element.
- **Double-remove idempotent:** `_remove_client()` is safe to call twice on the same socket.

9 Platform Integration

9.1 macOS

- **Dock hiding.** After GLFW initialization (in `_on_post_init()`), the display server sets `NSApplicationActivationPolicyAccessory` on the shared `NSApplication`. GLFW’s `glfwInit()` sets the policy to `Regular`; the post-init call overrides it. The Dock icon briefly flashes at launch. Requires `pyobjc-framework-Cocoa` (optional display extra).
- **Process name.** `setproctitle("Lux")` makes the process identifiable in `ps` and `top`. Does not affect Activity Monitor’s binary name column.
- **Font discovery.** Prefers Arial Unicode or Helvetica; merges Apple Symbols for mathematical/Z notation glyphs.
- **AF_UNIX path limit.** macOS limits Unix socket paths to ~104 characters. `tempfile.mkdtemp(prefix="lux-")` used in tests; production paths are short by design.

9.2 Linux

- **Font discovery.** DejaVu Sans or Noto Sans; merges Noto Sans Symbols2 for Unicode coverage.
- **XDG compliance.** Prefers `$XDG_RUNTIME_DIR/lux/` for socket placement.
- **No Dock equivalent.** Window managers use the X11 window title (set to “Lux”).

10 MCP Tool Surface

24 tools organized by function:

Tool	Signature and Purpose
Scene management	
<code>show</code>	(<code>scene_id</code> , <code>elements</code> , <code>title?</code> , <code>layout?</code> , <code>frame_id?</code> , <code>frame_title?</code> , <code>frame_size?</code> , <code>frame_flags?</code>) — Display a scene. Docstring lists all 24 element kinds.
<code>show_table</code>	(<code>scene_id</code> , <code>columns</code> , <code>rows</code> , <code>filters?</code> , <code>detail?</code> , <code>title?</code>) — Filterable data table with search, combo filters, and detail panel.
<code>show_dashboard</code>	(<code>scene_id</code> , <code>metrics?</code> , <code>charts?</code> , <code>table_columns?</code> , <code>table_rows?</code> , <code>title?</code>) — Metrics cards + charts + summary table.
<code>update</code>	(<code>scene_id</code> , <code>patches</code>) — Incremental JSON patches targeting elements by ID.
<code>clear</code>	() — Clear all scenes.
Communication	
<code>ping</code>	() — Latency probe; returns <code>PongMessage</code> .
<code>recv</code>	(<code>timeout?</code>) — Receive next event (interaction, window lifecycle).
<code>set_menu</code>	(<code>menus</code>) — Set agent-provided menu bar entries.
<code>register_tool</code>	(<code>name</code> , <code>description</code> , <code>schema</code>) — Register a custom tool in the display's tool palette.
<code>set_theme</code>	(<code>theme</code>) — Apply display theme.
Configuration	
<code>display_mode</code>	() — Read current display mode (y/n); advisory L3 signal for consumer plugins.
<code>set_display_mode</code>	(<code>mode</code>) — Set display mode.
<code>set_window_settings</code>	(<code>opacity?</code> , <code>font_scale?</code> , <code>decorated?</code> , <code>fps_idle?</code>) — Configure display window appearance and idle frame rate.
<code>set_frame_state</code>	(<code>frame_id</code> , <code>minimized?</code>) — Control frame state (minimize/restore).
Introspection	
<code>inspect_scene</code>	(<code>scene_id</code>) — Return element tree for a scene.
<code>list_scenes</code>	() — List all active scenes with metadata.
<code>screenshot</code>	() — Capture display as base64 PNG.
<code>get_display_info</code>	() — Display dimensions, frame count, client count.
<code>get_window_settings</code>	() — Current window geometry.
<code>get_theme</code>	() — Current theme name.
<code>list_clients</code>	() — Connected clients with names and scene counts.
<code>list_menus</code>	() — Registered menu items.
<code>list_recent_events</code>	(<code>count?</code>) — Recent interaction events.
<code>list_errors</code>	(<code>count?</code>) — Recent error log entries.

11 Formal Model

The display server’s state machine is formally specified in `docs/display-server.tex` using the Z notation (ISO 13568). The specification covers:

- Client connection/disconnection with FrameReader association.
- Scene replacement, incremental patching, and clearing.
- Interaction event generation and broadcast.
- FrameReader streaming buffer invariants.

The specification has been type-checked with Spivey’s `fuzz` and model-checked with ProB (7,943 reachable states explored). Refinement tests in `tests/test_display_refinement.py` verify commutativity between the abstract Z operations and the concrete Python implementation via an abstraction function (`tests/display_abstraction.py`).

Partition tests in `tests/test_display_partition.py` derive test cases from the Z specification using the Test Template Framework (TTF), covering all input space partitions of each operation schema.

Note: the formal model predates multi-scene tabs and models a single-scene display. The abstraction function maps multi-scene state to the active tab for backward compatibility. A specification update to model the full multi-scene state is pending.

12 Test Architecture

Test File	Count	What It Tests
<code>test_protocol.py</code>	168	Serialization, deserialization, framing
<code>test_display_partition.py</code>	107	TTF-derived partitions from Z spec
<code>test_tools.py</code>	83	MCP tool behavior (mocked client)
<code>test_display_state.py</code>	65	State machine logic (no ImGui)
<code>test_display_refinement.py</code>	28	Abstraction function commutativity
<code>test_display_client.py</code>	26	High-level client library
<code>test_query.py</code>	23	Generic introspection queries
<code>test_show.py</code>	22	CLI show subcommands
<code>test_runtime.py</code>	22	CodeExecutor lifecycle
<code>test_hooks.py</code>	18	Hook handler output
<code>test_config.py</code>	16	Config read/write
<code>test_paths.py</code>	16	Socket discovery, PID, auto-spawn
<code>test_remote.py</code>	12	mcp-proxy and remote transport
<code>test_cli.py</code>	11	CLI command smoke tests
<code>test_paths_hub.py</code>	11	Hub socket and PID paths
<code>test_service.py</code>	9	Daemon lifecycle
<code>test_unit.py</code>	6	Misc unit tests
<code>test_hub.py</code>	6	WebSocket session hub
<code>test_version.py</code>	2	Version string consistency
Marker-gated (not in default pytest run)		
<code>test_integration.py</code>	8	Socket IPC, protocol framing (real sockets)
<code>test_e2e.py</code>	3	Full display server lifecycle (requires GPU)
Total	662	(651 default + 11 marker-gated)

Quality gates: `ruff` (lint + format), `mypy` (strict), `pyright` (strict), `pytest` (all markers except `slow/integration/e2e`). Zero violations required.